Rain Model for Events

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1 Climatology

1.1 Estimation of the probability of rain

We start from the fact that rain exceeding a certain threshold t (for example 2 mm) on a given day D follows a Bernoulli distribution:

$$X_D \sim \text{Bernoulli}(p) \tag{1}$$

The time window we consider contains all the days within ± 7 days of the event day, over the past 30 years. In reality, consecutive days are not independent, since rainy days tend to come in series (if it rains the day before, the probability that it rains today is higher). Therefore, rather than considering each day as a realisation, we consider each year as a realisation. It means that we compute the observed proportion of rainy days above the threshold within the time window for one year. In other terms:

$$\hat{p}_y \;=\; \frac{k_y}{n_y} \;=\; \frac{1}{n_y} \sum_{d \in W_y} \mathbf{1}\{R_{y,d} \geq t\}, \qquad y = 1, \ldots, Y, \tag{2}$$

where W_y is the 15-day window around the event date in year y, $n_y = |W_y|$ is the number of available days in that window, $R_{y,d}$ is the rainfall on day d of year y, t is the rain threshold, and $Y = 30$ is the number of years.

$\hat{p}_y$ is therefore the estimator of p_y, the probability of observing an above the threshold rain in one day of the selected 15-day period, for the year y. This model does not assume the independence of the days within a given year.

We then compute the estimator $\hat{p}$ as the mean of those $\hat{p}_y$:

$$\hat{p} \;=\; \frac{1}{Y} \sum_{y=1}^{Y} \hat{p}_y, \tag{3}$$

which is the estimator of the probability of observing an above the threshold rain in one day of the selected 15-day period.

Considering such a time window first allows us to use much more data than only the exact same day of the previous years. Second, it gives us enough years to use the Central Limit Theorem (CLT) to compute confidence intervals.

1.2 Moments and CLT

We apply the CLT under the weaker assumption that the years are independent, while the days within the same year may be dependent. Assuming $\hat{p}_1, \ldots, \hat{p}_Y$ i.i.d. with mean p and variance

σ^2,

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{Var}(\hat{p}) = \frac{\sigma^2}{Y}, \qquad \hat{p} \overset{\text{approx.}}{\sim} \mathcal{N}\left(p, \frac{\sigma^2}{Y}\right) \text{ by the CLT.} \tag{4}$$

1.3 Confidence interval

Estimating σ^2 by $s^2 = \frac{1}{Y-1}\sum_y (\hat{p}_y - \hat{p})^2$, we obtain the 95% confidence interval

$$\mathrm{CI} = [\, p_-,\ p_+ \,] = \left[\, \hat{p} \pm t_{Y-1;\,0.975}\, \frac{s}{\sqrt{Y}} \,\right] \cap [0,1], \tag{5}$$

where $t_{Y-1;\,0.975} \approx 2.045$ is the 0.975-quantile of the Student distribution with $Y-1 = 29$ degrees of freedom. We intersect with $[0,1]$ since a probability cannot lie outside this range.

1.4 Validation (backtest)

Procedure. To check the method on real data, we replay it in the past. For every day i of the test years 2016–2025, in four locations (Zurich, Geneva, Lugano, Seville) and for two thresholds ($t = 2$ and 10 mm), we compute the estimate $\hat{p}_i$ of (3) using *only* the Y years preceding the test year, and record the outcome

$$o_i = \mathbf{1}\{R_i \geq t\}, \qquad i = 1, \ldots, N,$$

where R_i is the observed rainfall on day i. This gives about $N \approx 3\,650$ days per location and threshold. ERA5 data are used both for the estimation and for the outcomes.

Reliability. A probability forecast is *reliable* (or calibrated) if, among all days where it announces a probability close to p, rain is observed with frequency close to p. We split $[0,1]$ into classes $B_1, \ldots, B_K$ and, for each class, compare the mean announced probability with the observed frequency:

$$\bar{p}_k = \frac{1}{|I_k|}\sum_{i \in I_k} \hat{p}_i, \qquad \bar{o}_k = \frac{1}{|I_k|}\sum_{i \in I_k} o_i, \qquad I_k = \{\, i : \hat{p}_i \in B_k \,\}.$$

The forecast is reliable if $\bar{o}_k \approx \bar{p}_k$ for all k.

Class B_k	Days $\lvert I_k \rvert$	Announced $\bar{p}_k$	Observed $\bar{o}_k$
$[0.00, 0.05)$	2 953	0.019	0.023
$[0.05, 0.10)$	3 360	0.076	0.079
$[0.10, 0.15)$	6 115	0.127	0.119
$[0.15, 0.20)$	5 784	0.171	0.170
$[0.20, 0.30)$	1 789	0.238	0.268
$[0.30, 0.40)$	6 944	0.353	0.337
$[0.40, 0.50)$	2 251	0.435	0.403

Table 1: Reliability of the climatological estimate ($Y = 30$, all locations and thresholds).

In every class, the difference $|\bar{o}_k - \bar{p}_k|$ is at most about 3 percentage points: the probabilities given by the model are well calibrated.

2 Forecast

The problem with the climatology inference is that it only considers past events and **not** the current state of the atmosphere and the current weather. We therefore need to include the forecast, which is much more precise if the event is soon but loses its value after roughly 2 weeks. The main idea here is to trust climatology more if the event is far away, and to put more emphasis on the forecast if the event is close to the current date.

2.1 Weight function

We therefore consider a weight function $w(h)$, with h the number of days between today and the event, and define our final estimate $\tilde{p}$ as follows:

$$\tilde{p} = w(h)\, p_f + \big(1 - w(h)\big)\, \hat{p}, \qquad w(h) = \max\!\left(0,\ 1 - \frac{h}{16}\right) \tag{6}$$

with p_f the probability given by the forecast and $\hat{p}$ the probability given by the climatology (3). The number 16 comes from the fact that the API we use stops making forecasts after 16 days; it is not a statistical choice.

As we can see, we defined a linear weight for now; improving it is a possible improvement for the future. A good method could be to use the Brier score, choosing for each horizon h the weight that minimizes the squared error between the given probability and what really happened.

Another limitation of the model is that p_f represents the probability of any rain (even 0.1 mm of rain), whereas $\hat{p}$ represents the probability of rain exceeding the threshold t. As we want to interpret $\tilde{p}$ in (6) the same way as $\hat{p}$, we tend to overestimate $\tilde{p}$, because p_f drives $\tilde{p}$ higher than it should be. A correction using conditional probability could be a good idea:

$$\Pr(\text{rain} > t) = \Pr(\text{rain} > t \mid \text{rain} > 0)\ \Pr(\text{rain} > 0) \approx \hat{q}\, p_f, \qquad \hat{q} = \frac{k}{k_0},$$

with k the number of days with rain exceeding t and k_0 the number of days with any rain in the time window. We could then replace p_f by $\hat{q}\, p_f$ in (6).

2.2 Confidence interval

We decide to assume that p_f is deterministic and has no variance (meaning that the probability given by the API is the true probability). This is a strong assumption and is not expected to be true, but we make it for the sake of simplicity. Applying the weights of (6) to the bounds of (5), we compute the confidence interval as:

$$\mathrm{CI}_{\text{final}} = [\, p_{\text{lo}},\ p_{\text{hi}}\,] = w\, p_f + (1 - w)\, [\, p_-,\ p_+\,] \tag{7}$$

where $w = w(h)$ is the weight for the given horizon h. Note that for $h = 0$, we have $w = 1$, so $\mathrm{CI}_{\text{final}}$ reduces to the single point p_f: because of our assumption, the verdict (12) can then never be a close call.

3 Decision

3.1 Loss without protection

On the event day D, the rain is $X_D \sim \text{Bernoulli}(p)$, with p estimated by $\tilde{p}$ from (6), and the organizer's loss L is:

$$L = \lambda R \cdot X_D = \begin{cases} \lambda R & \text{with probability } p, \\ 0 & \text{with probability } 1 - p, \end{cases} \tag{8}$$

where R is the total revenue of the event and λ the share of the revenue lost in case of rain above the threshold t.

It means that the expected value of the loss is:

$$\mathbb{E}[L] = \lambda R\, p \tag{9}$$

3.2 Loss with protection

We suppose here that the protection is perfect, meaning that for a certain cost C, paid with certainty, the loss is brought back to 0.

3.3 Decision threshold

If the organizer compares the expected losses, they protect themselves when the protection is cheaper than the expected loss (9). In other words, when:

$$C < p\,\lambda R \quad \Longleftrightarrow \quad p > \frac{C}{\lambda R} \tag{10}$$

The threshold is then defined by:

$$\boxed{p^* = \frac{C}{\lambda R}} \tag{11}$$

which is a classic cost–loss model, with p^* being the cost–loss ratio.

3.4 Interpreting p^*

p^* is the price of the protection as a percentage of what we lose in case of a rainy day. The rule (10) is therefore pretty easy to understand: "protect yourself if the probability of rain exceeds the price of the protection divided by the potential loss."

Here an important assumption is made regarding the organizer: **risk neutrality**. This assumption is often not realistic, in particular for small organizers. Indeed, a potential loss of 12 000 CHF with a probability of 25 % may not be judged the same way as a certain loss of 3 000 CHF, because the 12 000 CHF loss could be too damaging. This could be improved by modeling the risk aversion of the organizer, or by adding a constraint on the maximum acceptable loss.

3.5 Dealing with the uncertainty of p

If p were known exactly, there would be only two cases: $p > p^*$ or $p < p^*$. But p is estimated by $\tilde{p}$, with the interval $\mathrm{CI}_{\mathrm{final}} = [p_{\mathrm{lo}}, p_{\mathrm{hi}}]$ from (7). Depending on where the threshold p^* (11) lies relative to this interval, the verdict is different:

$$\text{verdict} = \begin{cases} \textbf{Protect} & \text{if } p_{\mathrm{lo}} > p^* \quad \text{(the whole interval is above the threshold)} \\ \textbf{Don't protect} & \text{if } p_{\mathrm{hi}} < p^* \quad \text{(the whole interval is below the threshold)} \\ \textbf{Close call} & \text{if } p_{\mathrm{lo}} \leq p^* \leq p_{\mathrm{hi}} \quad \text{(the threshold is inside the interval)} \end{cases} \tag{12}$$

3.6 Other assumptions

Other assumptions have been made for the sake of simplicity, and they are all possible ways to improve our model. Statistics is not about finding the truth, but about getting as close as possible to it :)

Assumption	**Reality**
Independent, identically distributed years	The climate changes over 30 years, so p may not be the same for all years.
Binary rain (above or below t)	The loss depends on the intensity and duration of the rain.
Fixed λ	The loss also depends on when the rain falls (in the morning or during the concert).
Perfect protection	A tent does not save everything: some residual loss usually remains.
Single day	A three-day festival requires combining several days, which are correlated with each other.

Table 2: Assumptions of the model